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# SkyHopper PRO User Manual

## Version 1.44.10

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## Abbreviation List

| Abbreviation | Syntax                                                                    |
|--------------|---------------------------------------------------------------------------|
| ARQ          | Automatic Repeat reQuest                                                  |
| BPS          | Bits Per Second                                                           |
| BW           | Bandwidth                                                                 |
| CDD          | Cyclic Delay Diversity                                                    |
| CEP          | Circular Error Probable (Accuracy)                                        |
| CINR         | Carrier to Interference and Noise Ratio (Signal to Noise Ratio)           |
| CPLD         | Complex Programmable Logic Device                                         |
| CRC          | Cyclic Redundancy Code                                                    |
| CTC          | Convolutional Turbo Code                                                  |
| dB           | Decibel                                                                   |
| dBm          | Power ratio in decibels of the measured power referenced to one mill watt |
| DC           | Direct Current                                                            |
| DFS          | Dynamic Frequency Selection                                               |
| EMI          | Electromagnetic Interference                                              |
| ETH          | Ethernet                                                                  |
| ETSI         | European Telecommunication Standard Institute                             |
| FAE          | Field Application Engineer                                                |
| FCC          | Federal Communications Commission                                         |
| FEC          | Forward Error Correction                                                  |
| FFT          | Fast Fourier Transform                                                    |
| FPS          | Frame Per Second                                                          |
| Freq.        | Frequency                                                                 |
| FW           | Firmware                                                                  |
| GHz          | Gigahertz                                                                 |
| GPS          | Global Positioning System                                                 |
| HW           | Hardware                                                                  |
| IAS          | Interference Avoidance Solution                                           |
| ID           | Identification (Number)                                                   |
| IP           | Internet Protocol                                                         |
| LAN          | Local Area Network                                                        |
| LAT          | Latitude                                                                  |
| LED          | Light Emitting Diode                                                      |

|      |                                            |
|------|--------------------------------------------|
| LON  | Longitude                                  |
| LOS  | Line-Of-Sight                              |
| MHz  | Megahertz                                  |
| MIMO | Multiple-In-Multiple-Out                   |
| MRC  | Maximal Ratio Combining                    |
| MW   | Milliwatt                                  |
| N.C. | Not Connected                              |
| OEM  | Original Equipment Manufacturer            |
| OFDM | Orthogonal Frequency Division Modulation   |
| OS   | Operating System                           |
| OSD  | On Screen Display                          |
| PC   | Personal Computer                          |
| PPS  | Pulse Per Second                           |
| PTP  | Point-To-Point                             |
| PTMP | Point-To-Multipoint                        |
| QAM  | Quadrature Amplitude Modulation            |
| QPSK | Quadrature Phase Shift Keying (modulation) |
| RF   | Radio Frequency                            |
| RMS  | Root Mean Square (Average)                 |
| RSSI | Receiver Signal Strength Indication        |
| Rx   | Receive                                    |
| SBAS | Satellite Based Augmentation System        |
| SDR  | Software Defined Radio                     |
| SW   | Software                                   |
| TDD  | Time Division Duplexing                    |
| TDMA | Time Division Multiple Access              |
| Tx   | Transmit                                   |
| UAV  | Unmanned Aerial Vehicle                    |
| UGV  | Unmanned Ground Vehicle                    |
| VLAN | Virtual LAN                                |
| VPN  | Virtual Private Network                    |

**Table 1: Abbreviation List**

## 1. SkyHopper PRO User Manual

Thank you for choosing SkyHopper PRO, a robust data link designed for drones and robotics. This user manual explains how to operate the SkyHopper PRO.

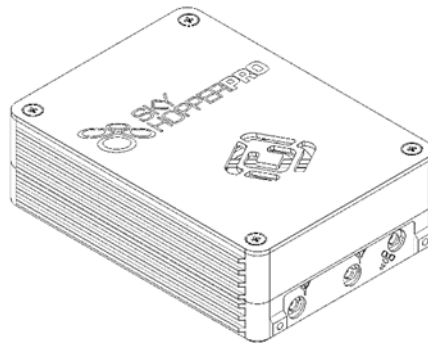
## 2. Overview

SkyHopper PRO combines advanced wireless technologies for long-range and Non-Line of Sight (N-LOS) communication. It supports broadcast, multicast, and unicast transmission modes. Depending on the version, it can handle Point-to-Point and Point-to-Multipoint communication, enabling operations like multi-drone control and communication between drones and multiple ground units.

SkyHopper PRO integrates control, telemetry, and payload data into a single RF channel. Built on the Software Defined Radio (SDR) concept, it features a sophisticated FPGA and a dual-core processor, ensuring seamless hardware and software integration.

The programmable RF section with a cutting-edge RFIC allows flexibility across various communication protocols and environments. This makes the system adaptable for both commercial and defense applications.

Mobilicom's SDR is designed for UAVs and robotics, providing high performance in a compact, lightweight design. It uses OFDM - TDD technology to enhance spectral efficiency and robust communication. The system can support both Mesh networks and Point-to-Point configurations, optimizing performance based on mission needs. Additionally, it includes advanced cybersecurity features to ensure secure communication in harsh or hostile environments.



**Figure 1: Sky Hopper PRO**

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### 3. Main Features

SkyHopper PRO offers the following features:

- Non-Line-of-Sight (NLOS) Communication: Utilizes proprietary technology for robust and clear signals.
- Robust Security: Advanced encryption mechanisms to protect data from cyberattacks.
- Dynamic Frequency Selection: Adapts to the best available frequencies.
- Interference Avoidance: Advance feature enabling interference avoidance (Requires ICE Professional license).
- Software Defined Radio (SDR): Mobile technology for flexible and programmable radio functionality.
- 2-Channel MIMO: Full support for Multiple Input Multiple Output.
- Compact and Lightweight: Small physical dimensions with low weight.
- Energy-efficient design.

### 4. Packets Supported by the SkyHopper PRO Network

The SkyHopper PRO physical link supports Ethernet (IEEE 802.3), including TCP/IP networking and real-time applications using protocols like RTSP. It supports broadcast, multicast, and unicast transmissions. Any Ethernet packet received by the unit's Ethernet port is internally encapsulated with a standard frame for transmission via the Data Link to other units. The other unit then recovers the original Ethernet packet and forwards it to its Ethernet port.

SkyHopper PRO functions as an Ethernet layer 2 (MAC layer) switch, ensuring that integration with your equipment is as simple as connecting an Ethernet cable. This seamless compatibility eliminates the need for additional configuration or setup, making it easy to incorporate into your existing network infrastructure.

## 5. Point-to-Point (PTP)

A point-to-point system consists of two units, with a predefined bandwidth ratio between them to suit robotic and drone applications. The data link is asymmetric, allowing up to 10% of the bandwidth to one unit and 90% to the other. For example, in a drone scenario, the drone transmits video to the ground unit while the ground unit controls the drone's camera.

For a 6 Mbps configuration, 10% of the available bandwidth is allocated for uplink, and 90% is allocated for downlink.

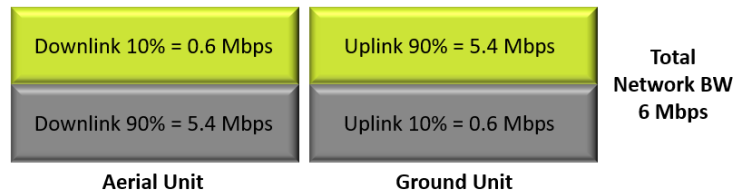


Figure 2: Asymmetric link configuration

## 6. Physical Connectors

Front View Connectors:

- RF1: Antenna 1 Connector, MMCX Female 50 Ohm
- RF2: Antenna 2 Connector, MMCX Female 50 Ohm
- GPS: GPS Antenna Connector, MMCX Female

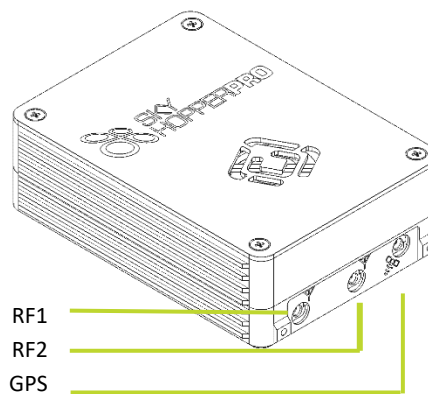
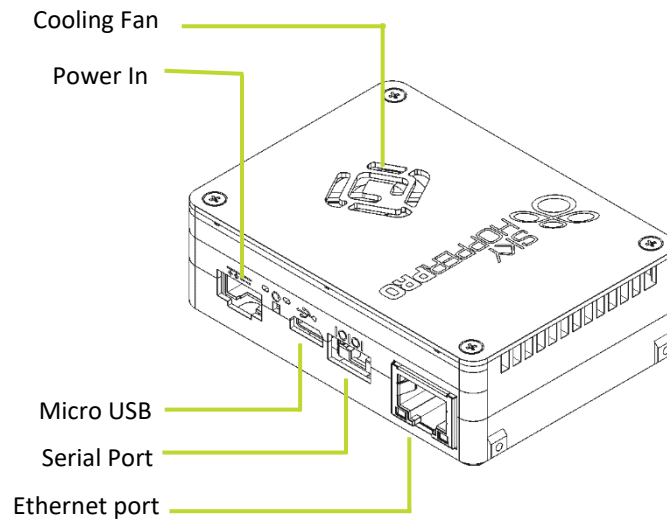


Figure 3: SkyHopper PRO Front View

#### Side View Connectors:

- Ethernet: Standard RJ45 connector
- Serial (IOIOI): Molex - 502380-0600
- RS232 / 1PPS connector
- Micro USB
- Power: ZE - ZER-04V-S
- Power connectors utilizing 7.4-24VDC



**Figure 4: SkyHopper PRO Connector Side view**

## 7. Technical Specifications

### Radio & Modem Specifications

| Radio and Modem         |                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| Range                   | Up to 5km / 3.1 miles per hop (LOS, uninterrupted conditions)<br>For longer ranges, other products available |
| Frequency               | 2.3-2.7 GHz (ISM) Or 5.8 GHz (ISM)<br>Other frequencies available upon request                               |
| Antenna's Connectors    | MMCX/F (Female) (x2)                                                                                         |
| Sensitivity             | Max -101dBm                                                                                                  |
| Transmission Technology | OFDM - TDD                                                                                                   |
| Configuration           | MIMO 2X2                                                                                                     |
| Channel BW              | 2.5MHz / 4.2MHz / 8.4MHz                                                                                     |
| Frequency Resolution    | 0.25MHz                                                                                                      |
| Diversity Support       | TX & RX Diversity                                                                                            |
| Network Topologies      | Point to – Point,<br>Point-to-Multipoint/Mesh, Relay (Repeater)                                              |
| Transmission Methods    | Broadcast, Multicast, Unicast                                                                                |
| Modulation              | QPSK                                                                                                         |
| Throughput (Bit Rate)   | Configurable from 0.95-6.4Mbps<br>For extended throughput, other products available                          |
| Encryption              | 128 AES – Standard Package<br>256 AES – Requires ICE Professional license                                    |
| Environmental           |                                                                                                              |

|                                     |                                                                                 |                 |
|-------------------------------------|---------------------------------------------------------------------------------|-----------------|
| Temperature Range                   | Operating                                                                       | Storage         |
|                                     | -20°C to 45°C                                                                   | -50°C to 85°C   |
| Humidity                            | 5% to 95%                                                                       |                 |
| Interfaces                          |                                                                                 |                 |
| Data                                | Ethernet, RS232                                                                 |                 |
| Full Ethernet Protocol Transparency | IP, TCP, UDP, RTP, HTTP<br>Broadcast, Multicast, Unicast<br>Ethernet, VLAN, VPN |                 |
| Telemetry & Control                 | RS232                                                                           |                 |
| Dimensions                          | [HxWxD] mm                                                                      | [HxWxD] inch    |
| PRO Casing Model                    | 26 x 54 x 74                                                                    | 1.0 x 2.1 x 2.9 |
| Weight                              |                                                                                 |                 |
| SkyHopper PRO                       | 105 Grams                                                                       |                 |
| Power                               |                                                                                 |                 |
| DC Voltage                          | PRO & PRO Lite: 7.4-24 V DC<br><br>PRO Micro: 6–20 V DC                         |                 |
| Power Consumption                   | 10W Average 12W Peak (@23dBm)                                                   |                 |
| Configuration and Management        |                                                                                 |                 |
| Monitoring and Configuration        | Web-Based Application                                                           |                 |
| Programmer’s Interface              | API                                                                             |                 |

**Table 2: Technical Specifications**

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## 8. Frequency Bands Available

**Frequency Bands Available:** SkyHopper PRO operates within the frequency bands of 2.3 - 2.7GHz or 4.9 – 5.9GHz, with the option for other bands upon request.

**Antennas:** The choice of antenna depends on the specific application and requirements. Different types of antennas, including directional or omnidirectional, may be more suitable for certain applications.

## 9. RF Characteristics

- **Peak output power:** 1W per channel (2W total)
- **Average output power:** 2 x 100mW per channel (@23dBm)
- **Noise Figure:** 5dBm
- **Transmission Technology:** OFDM - TDD
- **Diversity Support:** MRC -RX and CDD - TX
- **Frequency Resolution:** 0.25MHz
- **Channel bandwidth:** Configurable 2.5/4.2/8.4MHz
- **Modulation:** QPSK
- **FEC methods:** CTC
- **FEC Ratios:** 2/3, 1/2, 1/4 (According to the chosen bandwidth)
- **Error Detection:** CRC32
- **Full TDD asymmetrical duplexing**
- **Constant Ratio:** 9:1 (Remote: Controller)

## 10. GPS Capability

SkyHopper PRO features a built-in GPS receiver. In a Point-to-Point (PTP) network configuration, the GPS receiver provides location information, accessible to the customer through the SkyHopper API (available upon request). In a multi-unit network setup, the GPS receiver facilitates synchronization and can also be utilized for location purposes.

Active GPS antennas are utilized by SkyHopper PRO. Power for the active antenna is supplied by the unit through the standard GPS MMCX connector.

| Item            | Description          |
|-----------------|----------------------|
| Receiver Type   | GPS L1 C/A code      |
| Accuracy        | Position 2.5 m CEP   |
|                 | SBAS 2.0 m CEP       |
| Acquisition     | Cold start: 27 s     |
|                 | Aided start: < 3 s   |
|                 | Hot start: 1 s       |
| Sensitivity     | Tracking: -161 dBm   |
|                 | Cold start: -147 dBm |
|                 | Hit start: -156 dBm  |
| Timing Accuracy | RMS 30 ns            |
|                 | 99% < 60 ns          |
|                 | Granularity 21 ns    |

**Table 3: Embedded GPS Characteristics**

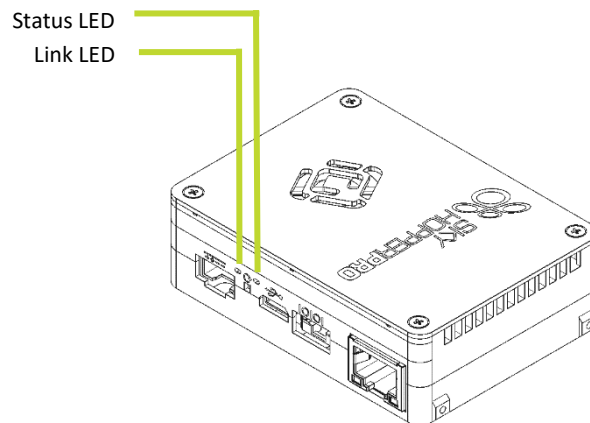
## 11. SkyHopper PRO LED Indicators

The SkyHopper PRO LED indicators provide information regarding the power and link status of the unit. The Status LED indicates the unit's operational status, while the Link LED indicates the status of the Ethernet link. These LEDs operate in four different modes: Off, On, Slow Blinking, and Fast Blinking.

During the unit's initialization, both LEDs remain constantly lit. When the Status LED is Off, it signifies that the unit is powered down. When it is On, it indicates that the unit is booting up. A Slow Blinking Status LED indicates that the remote unit is transmitting, and the controller unit is receiving during a Point-to-Point (P2P) search. A Fast-Blinking Status LED indicates that the air link is established.

The Link LED indicates the status of the Ethernet link. When off, it indicates no Ethernet connection is detected. When on, it signifies that the Ethernet cable is connected to a working device on the host, with no data transmission occurring. A Slow Blinking Link LED indicates an error in the Ethernet connection. A Fast-Blinking Link LED indicates active Tx and Rx air traffic.

In case of an error in the SkyHopper PRO unit, both the Status and Link LEDs will either simultaneously slow blink or remain constantly lit without transitioning into any blinking modes.



**Figure 5: SkyHopper PRO LED indicators**

## 12. Configuring a point-to-point Link

To configure a Point-to-Point (PTP) radio link between two SkyHopper PRO units, both units must have the same frequency and profile configuration. It is mandatory that the software (SW) version on both units is identical.

The units should have identical configurations, except for units IP and the designation of one as a Remote Unit and the other as a Controller Unit.

Upon power-up, the Remote Unit will begin transmitting regardless of the link status. Meanwhile, the Controller Unit will remain in Rx mode, actively searching for a preamble (synchronization pattern) to establish synchronization. Once the Controller Unit detects the preamble, it commences transmission according to its allocated time slot.

It's crucial to note that the transmission bandwidth ratio is predefined as part of the profile configuration, allowing for asymmetry of up to 10% to one unit (either remote or controller) and 90% to the other unit.

Additionally, a Viewer Unit functions similarly to a Controller Unit but only receives data and does not transmit.

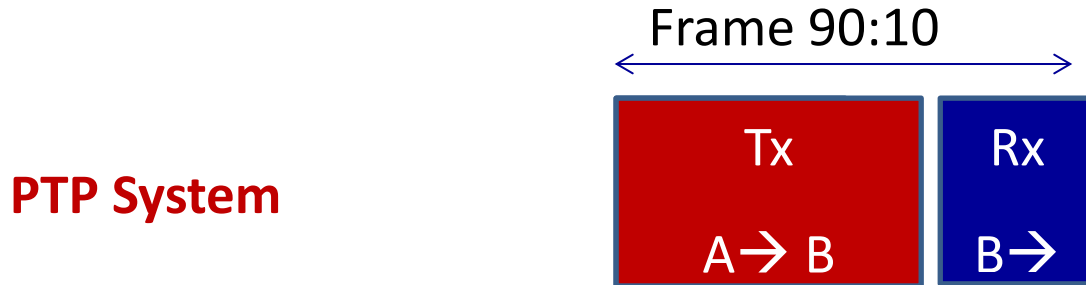


Figure 6: Transmit frames of the remote unit and controller unit.

## 13. PTP Link Step-by-Step Configuration

**CAUTION:** Before powering up the SkyHopper PRO units, antennas should be connected to RF1 and RF2 connectors to prevent irreparable damage to the unit.

To set up a Point-to-Point (PTP) link, you'll need two SkyHopper PRO units. One unit will serve as the Remote Unit, while the other will function as the Controller Unit. The units are shipped pre-configured as a linked pair. In the following steps, you will find how to set the most basic RF link in case the radios were returned to their defaults.

### 1. Connecting to the Unit:

- Begin by connecting one of the units to your PC using a standard Cat5 or higher Ethernet cable.
- Configure a static IP on your PC within the 192.168.131.xxx subnet.
- Open a web browser on your PC or laptop (we recommend Chrome) and enter the following address in the address bar: <https://192.168.131.241> to access the Web GUI interface (of the remote unit).

**Remark:** The default IP address of the Remote Unit is 192.168.131.241 and the default IP address of the Controller Unit is 192.168.131.242. These addresses can be changed according to customer requirements.

### 2. Accessing the Radio:

- It is also possible to access the radio using a default IP that is hardcoded in the software and remains unchanged even after a system reset.
- This default IP address is 10.56.57.58 and is identical across all SkyHopper units.
- To connect using this IP, ensure your PC's network settings are configured to be compatible with the 10.56.57.xxx subnet.

### 3. Configuration:

- Navigate to the "Basic Configuration" section and verify that the unit mode is set to Remote. Once confirmed, you can set the frequency you intend to use for this link.
- Go to the "Advanced Configuration" section and set the TX power that you would like to work with. If you don't know the needed power, set the TX power to High (default value).
- Select the RF bandwidth (RF BW) that you would like to work with and the relevant bitrate.
- These settings are the most basic configurations that you will need to establish a link. Please verify that you use the apply button after any change to confirm the settings.
- Once you have configured the Remote Unit, connect to the second unit, and change the unit mode to Controller. At this stage, the unit will perform a reset.
- Please reconnect to the unit and note that the IP address has been changed to 192.168.131.242.
- Once you reconnect to the Controller Unit, you must set the same parameters for frequency, RF BW, and bitrate, and apply changes after each modification. It is highly recommended to set both units with same Tx power.
- Once all the parameters are identical (except the unit type), the link should be established.

- You can verify the status of the link in the "General Information" section. The link status lines will turn green, and the link uptime will start counting.

#### 4. Additional Features:

Please note that SkyHopper PRO units offer additional features including DFS, pairing, sense, and avoid. Detailed explanations and configuration for these features can be found in their respective tabs.

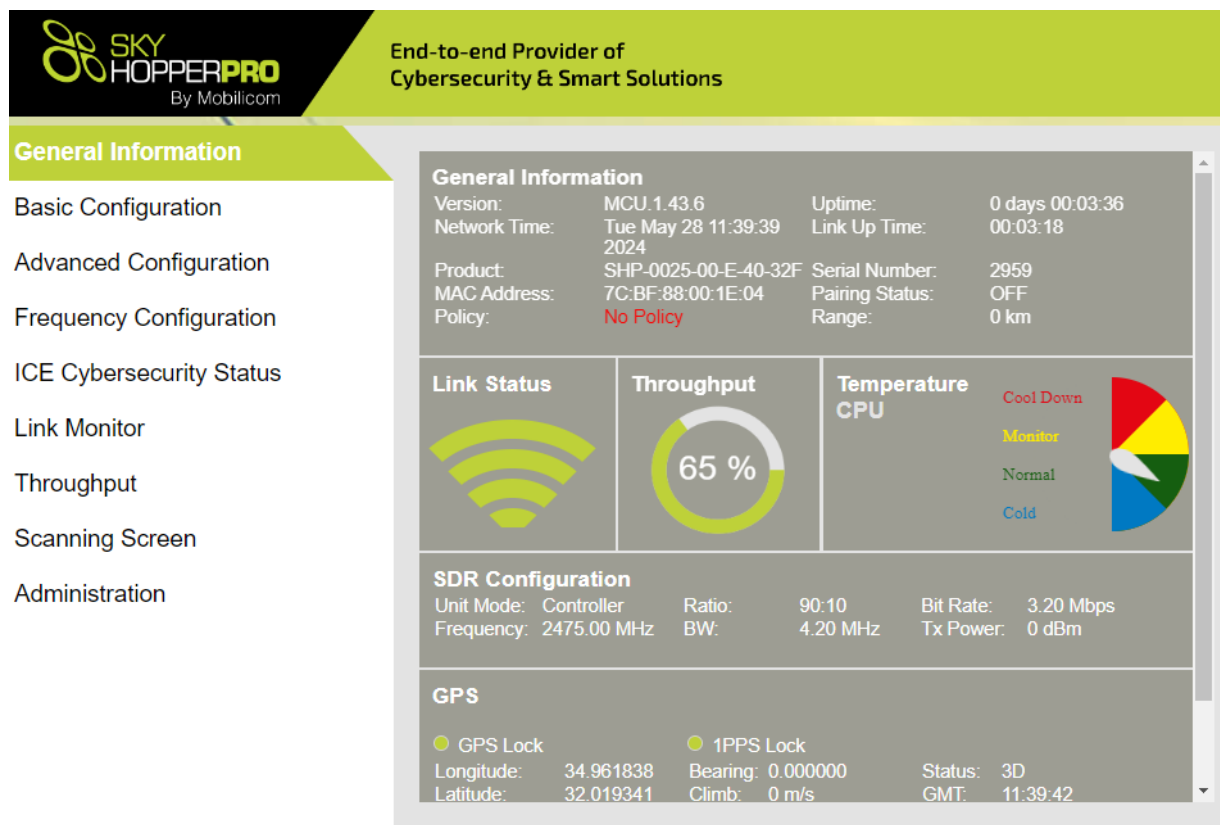


Figure 7: Web GUI General Information Page



End-to-end Provider of  
Cybersecurity & Smart Solutions

General Information

Basic Configuration

Advanced Configuration

Frequency Configuration

ICE Cybersecurity Status

Link Monitor

Throughput

Scanning Screen

Administration

**Unit Mode**

Remote
Controller
Viewer

**Frequency** Frequency (MHz)

**Unit IP**

**Subnet Mask**

**Default Gateway**

**Encryption**

Disable ☒ Enable ☐

AES Key ☒ 128 ☐ 256

Password

Erase Key

**Language(Language)**  ▼

**Time**

Time Zone  ▼

Time

Set PC Time

System Time Method

Manual ☒ Automatic ☐

SDR

☒ User Defined

☐ GPS

Refresh
Apply

**Figure 8: Web GUI “Basic configuration” Page**



End-to-end Provider of  
Cybersecurity & Smart Solutions

General Information

Basic Configuration

Advanced Configuration

Frequency Configuration

ICE Cybersecurity Status

Link Monitor

Throughput

Scanning Screen

Administration

**Units of Measurement**  
Imperial ☒ Metric

**Tx Power (dBm)**    ☐ High    ☒ Low

**RF BW**                    ☐ 4.2 MHz    ☒ 8.4 MHz

**Bit Rate**                    ☒ 6.4 Mb

**Reinforced Uplink**  
Disable ☒ Enable

**RS232**  

RS232 To RS232

57600

Mavlink

☐ Save GPS, Unit and Throughput Info

Refresh    Apply

00:00:38 : Setting - RS232 To RS232 is changed to 4200000 Hz  
00:00:38 : Setting - RS232 To Eth (on VC) is changed to 3200000 Mb  
00:00:38 : Setting - Dynamic Frequency Selection is changed to 2475.00  
MHz  
00:00:38 : Setting - Dynamic Frequency Selection is changed to 2475.00  
MHz

Save Local Log

Download Unit Log

Download Config Files

Figure 9: Web GUI “Advanced Configuration” Page

## 14. General Information Page

The General Information Page serves as the primary interface upon establishing a connection with the SkyHopper PRO. It provides vital information concerning the unit's operational status, configurations, and settings. Detailed explanations of each section follow below.

### 1. General Section:

- This section shows the unit's firmware version, serial number, MAC address, and product number.
- It also displays Network Time, Uptime (how long the unit has been running) and Link Up Time (how long the connection has been established).
- Pairing Status Indicates whether the pairing feature is enabled or disabled
- Current policy and range between units are also shown here.

### 2. Indicators Section:

- Here, you'll find indicators showing the link status between units. Green means a strong data link, Yellow means fair, Red means poor, and Grey means no connection.
- The section also shows the throughput percentage of the configured bitrate.
- Unit temperature is displayed with colors: blue for cold, green for normal, yellow for monitoring, and red for cooldown. If it reaches the red zone, turn off the unit until it cools down.
  - Green limit high = 75 degrees Celsius
  - Yello limit high = 84 degrees Celsius

### 3. SDR configuration & RF Settings:

- This section summarizes configuration & RF settings, including the unit's mode (Controller, Remote, or Viewer).
- It shows the channel throughput division ratio between the Remote unit and the Controller unit (always 90:10).
- It shows the Frequency, BW, Bite Rate and the TX power at which the unit operates in SDR mode is also displayed here.

### 4. GPS Section:

- Here, you can check the status of the GPS antenna/source connection.
- When a GPS signal is received, 'GPS Lock' turns green, followed by '1PPS Lock.'

Please note that it may take a few seconds for the GPS signal to be detected and for the coordinates to display.

- Green indicates a received GPS signal, while Grey means no signal.

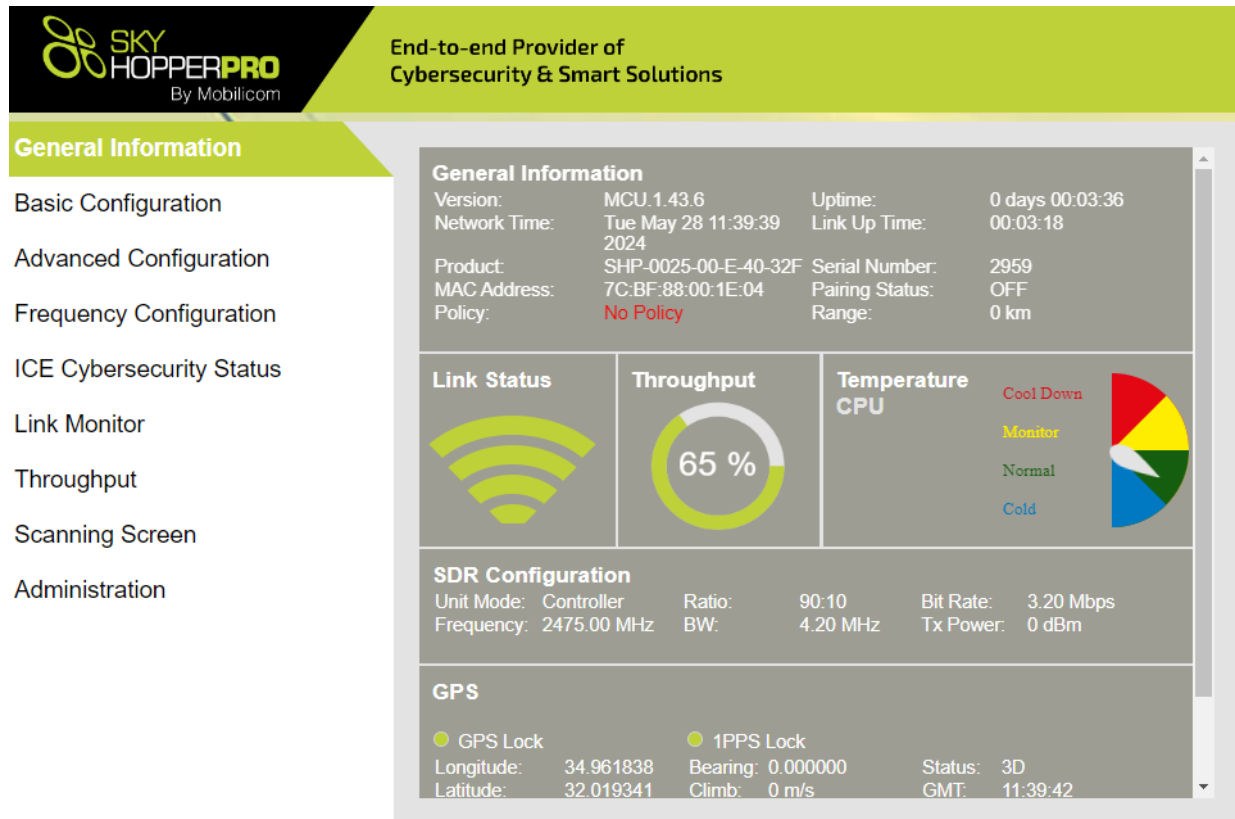


Figure 10: Web GUI General Information Page

## 15. Basic Configuration

The Basic Configuration screen provides essential settings for your SkyHopper PRO system. Here's an overview of the available options:

1. **Unit Mode:** Choose between "Remote", "Controller", or "Viewer" based on the unit's role in the Point-to-Point (PTP) link. The Remote Unit typically serves as the aerial unit, while the Controller Unit is the ground unit. A Viewer only receives data from the Remote Unit and doesn't transmit any data.
2. **Frequency:** Select either "Manual" or "DFS". In Manual mode, you can manually configure the desired frequency. In DFS mode, the system scans the frequency band and locks onto the best available channel. The frequency change resolution is 0.25 MHz.
3. **Unit IP:** Set the unit's IP to a value other than the default. Default IPs are: Remote unit - 192.168.131.241, Controller unit - 192.168.131.242, Viewer unit - 192.168.131.243. Configure Subnet Mask and Default Gateway if necessary for accessing the WEB GUI through routers or gateways.
4. **Subnet Task:** This feature allows you to configure subnet tasks for specific network operations.
5. **Default Gateway:** This setting allows you to configure the default gateway for network communication.
6. **Encryption:** Enable or disable Link AES encryption. If encryption is enabled, enter an AES encryption key in the password section. For AES 128 encryption, the key must be at least 8 characters long, and for AES 256 encryption, the key must be at least 9-32 characters long. Note that if only one unit has encryption enabled or if the keys do not match, the link will be established but no data will be transmitted.
7. **Language:** The Language tab in the GUI allows you to change the language of the interface using the configuration file. For advanced language support, please contact us for assistance at [support@mobilicom.com](mailto:support@mobilicom.com)
8. **System Time:** This setting allows you to specify how the system time is set. In manual mode, you can set the time based on user-defined input. In automatic mode, the system prioritizes time sources. If GPS is available, the time will be obtained from the GPS. Otherwise, the time will be set based on user-defined settings. Additionally, in manual mode, you can set the time manually or use the "Set PC Time" option. Once used, the unit will synchronize its time with the connected PC's time. Note that setting the unit time is not mandatory for a link to be established, but the unit's time will be reset to 00:00 on January 1st, 1970, if the unit is powered down or reset.

It's crucial to configure these settings accurately to ensure the proper functionality of your SkyHopper PRO system.



SKY  
HOPPERPRO  
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**Unit Mode**

Remote
Controller
Viewer

**Frequency** Frequency (MHz)

**Unit IP**

**Subnet Mask**

**Default Gateway**

**Encryption**

Disable ☒ Enable ☐

AES Key ☒ 128 ☐ 256

Password

Erase Key

**Language(Language)**  ▼

**Time**

Time Zone  ▼

Time

Set PC Time

System Time Method

Manual ☒ Automatic ☐

SDR

☐ User Defined

☒ GPS

Refresh
Apply

Figure 11: Web GUI “Basic Configuration” screen

## 16. Advanced Configuration

The Advanced Configuration screen empowers users to fine-tune their SkyHopper PRO system. Here are the configuration options available:

1. Units of Measurement: Choose between Imperial or Metric units of measurement.
2. Tx Power (dBm): Select between High, Low, or manually set (-45dBm to 23dBm) the transmitting power. High is the maximum power of the unit (23dBm). Low is 0 dBm, suitable for lab use or short distances (up to 10s of meters).
3. Channel Bandwidth: Specify the channel bandwidth in MHz, which affects the allowed bit rates.
4. Bit Rate: Choose the bit rate in Mbps. Lower bit rates enhance RF link performance, influenced by channel bandwidth and coding rate.
5. Reinforced Uplink: Enables meticulous configuration of crucial parameters including frequency allocation, transmission power levels, channel bandwidth, Forward Error Correction (FEC) rate, and more. Reinforced Uplink can be enabled when necessary.
6. Uplink Segment Attention: In scenarios involving aerial platforms and robotics, the uplink segment, referring to data transmission from ground station to aerial unit, requires special attention. Although often low bandwidth, it demands high resilience, surpassing downlink importance.
7. Automatic Repeat Request (ARQ): The **ARQ (Automatic Repeat Request)** feature enhances data reliability by automatically retransmitting lost or corrupted data packets. This ensures consistent and secure communication, particularly in challenging network environments.

**Feature Activation:** The ARQ feature is now included as a standard capability and does not require a license key. It can be activated directly through the Web GUI.

**Bandwidth Considerations:** When ARQ is activated, it utilizes up to **10% of the available bandwidth** to ensure the retransmission of packets, optimizing data integrity.

**Configuration Options:** The ARQ feature offers flexible configuration per interface and direction:

**Interface Support:** Ethernet, Serial

### Direction Settings

- Uplink: Data retransmission from the device to the network.
- Downlink: Data retransmission from the network to the device.
- Bidirectional: Enables retransmission for both uplink and downlink.

### Strategy Settings

The ARQ (Automatic Repeat reQuest) feature allows you to select the desired operational strategy based on your system requirements.

When ARQ is activated, it utilizes up to 10% of the available bandwidth to ensure the retransmission of packets, optimizing data integrity.

The utilization of the bandwidth is determined by the selection of the policy.

- **Basic (3% utilization):** Minimal retransmission, suitable for low-latency needs.
- **Mild (7% utilization):** Balanced retransmission, offering moderate reliability and bandwidth efficiency.
- **Advanced (10% utilization):** Maximum retransmission, designed for high reliability.

**Note: The ARQ settings must be identical on both the remote unit and the controller. If the configurations do not match, the RF link will fail to establish.**

8. RS232 Configuration: Configure serial data as disabled, RS232-To-RS232, RS232-to-Ethernet (Ethernet on Remote side), or RS232-To-Ethernet (Ethernet on Controller side). Exact settings must match on both sides for serial data flow. "RS232-To-Ethernet" enables serial data frame encapsulation and transfer over UDP/IP, applicable to Remote or Controller Units, but not simultaneously on both sides. The RS232 physical port is user-selectable on both sides.
9. Bit Rate Configurations: Choose bit rate configuration between 4800-921600 bps.
10. Serial Protocol Configurations: Select from UAV Navigation, MicroPilot, MavLink, and Other (generic RS232). RS232 packet length (bytes), start-byte, and end-byte vary based on selected protocol.

| Protocol Type  | RS232 Packet Total Length (bytes) | Start-Byte                 | End-Byte |
|----------------|-----------------------------------|----------------------------|----------|
| UAV Navigation | 71                                | 0xAC13                     | Any      |
| Micro Pilot    | Any, up to 1200                   | 0xFE                       | 0xFF     |
| MavLink        | MavLink 1 or MavLink 2(Pixhawk)   | (v1.0: 0xFE)<br>(v2: 0xFD) | Any      |
| Other          | Generic RS32 protocol             |                            |          |

**Table 4: Serial Protocol Configurations**



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**Units of Measurement**  
Imperial ☒ Metric

**Tx Power (dBm)**    ● High    ● Low    ● 0

**RF BW**                    ● 4.2 MHz    ● 8.4 MHz

**Bit Rate**                    ● 6.4 Mb

**Reinforced Uplink**  
Disable ☒ Enable

**RS232**  

RS232 To RS232 ▼

57600 ▼

Mavlink ▼

☐ Save GPS, Unit and Throughput Info

Refresh Apply

```
00:00:38 : Setting - RS232 To RS232 is changed to 4200000 Hz
00:00:38 : Setting - RS232 To Eth (on VC) is changed to 3200000 Mb
00:00:38 : Setting - Dynamic Frequency Selection is changed to 2475.00
MHz
00:00:38 : Setting - Dynamic Frequency Selection is changed to 2475.00
MHz
```

Save Local Log

Download Unit Log

Download Config Files

Figure 12: Web GUI “Advanced Configuration” screen 1



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Disable ☒ Enable

### RS232

RS232 To RS232

57600

Mavlink

|               |              |         |       |
|---------------|--------------|---------|-------|
| Ethernet Side | 192.168.0.11 | IP Port | 50111 |
| RS232 Side    | 192.168.0.14 | IP Port | 50101 |

### ARQ

Disable ☒ Enable

#### Ethernet

Disable ☒ Enable

#### Direction

☐ Uplink
☒ Downlink
☐ Bidirectional

#### Strategy

☐ Basic
☒ Mild
☐ Advanced

#### Serial

Disable ☒ Enable

☐ Downlink
☒ Bidirectional
☐ Advanced

☐ Save GPS, Unit and Throughput Info
 

Refresh Apply

00:00:31 : Setting - Subnet Mask is changed to 255.255.255.0  
 00:00:31 : Setting - Default Gateway is changed to N/A  
 00:00:31 : Setting - Dynamic Frequency Selection is changed to 2475.00 MHz  
 00:00:31 : Setting - Dynamic Frequency Selection is changed to 2475.00 MHz  
 00:00:31 : Setting - RS232 mode is changed to RS232 To RS232

Save Local Log

Download Unit Log

Download Config Files

Figure 13: Web GUI “Advanced Configuration” screen 2

## 17. Frequency Configuration & Interference Avoidance

The SkyHopper PRO includes a comprehensive frequency-management mechanism designed to ensure robust, interference-resistant, and secure wireless communication.

These functions combine **Pairing**, **Transmission Security (TranSec)**, **Dynamic Frequency Selection (DFS)**, and the **Interference Avoidance System (IAS)**, including **Sense and Avoid** capabilities.

Together they maintain optimal link performance by continuously monitoring the RF environment and adapting operation to the cleanest available channel.

### 1. Pairing

Pairing allows two or more units to participate in the same secured network. Units sharing the same pairing key can interoperate and exchange DFS/IAS data.

- **Manual Pairing:** The operator defines an 8-character password that must be entered on both units.
- **Auto Pairing:** Once both remote and controller units have an established RF link, enabling Auto Pairing automatically generates and distributes a secure key to both sides.

**Important:** Both units must share the same pairing key; otherwise, an RF link cannot be established. Pairing must be active for DFS and IAS features to operate.

### 2. TranSec

**Transmission Security (TranSec)** safeguards the air interface from interception by randomizing the pilot sequence according to the pairing key.

TranSec 2 and 3 modes (available under an ICE Professional license) apply dynamic pilot randomization to ensure that only paired units can synchronize on the same frequency.

### 3. Dynamic Frequency Selection (DFS)

DFS enables the radio to automatically locate and use the cleanest channel available.

During initialization, each unit scans a predefined range or list of frequencies, measures interference levels, and selects the one with the lowest average noise.

### 4. DFS Operation

- **Remote Unit:** Performs a complete frequency scan and begins transmitting on the channel with minimal interference.  
If a link is not established within the defined period, it restarts the scan automatically.
- **Controller Unit:** Searches the same range, detects the remote's signal preamble, and aligns its receiver. Once synchronized, it reports the selected frequency to the remote for future reference.

DFS can operate in two modes:

- **Range DFS:** The operator defines a start and end frequency for scanning.
- **Bank DFS:** The operator defines a table of discrete frequencies; the unit selects the best from this list.

**Use Case Example:**

When operating in an urban area with heavy Wi-Fi traffic, enabling DFS allows the SkyHopper PRO to automatically avoid occupied Wi-Fi channels in the 2.4 GHz ISM band, maintaining continuous video and telemetry streams without manual retuning.

## **5. Interference Avoidance System (IAS)**

IAS continuously evaluates link quality through the Packet Error Rate (PER).

If the PER exceeds a predefined threshold for a defined duration, the system automatically triggers a frequency switch to a cleaner channel.

**IAS uses two parameters:**

- **Policy:** Determines sensitivity to interference (see Table 5 for the PER scale).
- **Time Period:** Defines how long the PER must exceed the threshold before a change occurs.

The controller and remote units coordinate the switch through their management link to ensure simultaneous transition and minimal downtime.

**Use Case Example:**

If a nearby transmitter temporarily occupies the current channel, the PER increases.

When the PER exceeds the policy limit (for example 4 % under Policy 8) for 5 seconds, both units automatically migrate to the next best frequency, maintaining control and video continuity.

## **6. Sense and Avoid Capabilities (requires ICE Professional license)**

Sense and Avoid extends IAS functionality by actively monitoring interference and dynamically selecting new channels when degradation is detected.

- **Policy:** Defines the acceptable error rate.  
Values 1 to 15 correspond to PER thresholds from 0.5 % to 30 %.
- **Time Period:** Defines the reaction time (default 5 s).
- **Operation:** When the measured PER exceeds the defined threshold for the specified period, the system scans its configured list, compares signal quality, and switches to the channel with the best conditions.

The system uses the node and controller scan results to coordinate the switch, minimizing link interruptions.

**Example:**

With Policy 8 ( $\approx 4\%$  PER) and Time Period = 5 s, the units tolerate minor transient errors but will relocate the link if interference persists for more than five seconds.

**7. Aging Node Information**

After each DFS or IAS cycle, the node transfers its scan results to the controller, which stores them for future calculations.

To ensure decisions are based on current conditions, stored data is automatically aged and removed after a predefined interval.

- Default Time: 300 seconds
- Effect: Expired nodes are marked with  $-120$  dBm RSSI for both channels.
- Configuration: Can be enabled or disabled via configuration file; the timer value is fixed.

This prevents outdated RSSI measurements from influencing subsequent frequency selections.

**8. Blacklist Cleaning**

During DFS or IAS operation, channels that fail selection are temporarily placed on a blacklist.

After the link remains stable, the system automatically clears this list to allow reevaluation of previously excluded channels.

- **Default Timer:** 30 seconds after link-up
- **Condition:** Clearing occurs only after the node has reported its scanning data.

This guarantees that channels are periodically reconsidered and the system remains adaptive to changing RF conditions.

## 9. Enhanced Frequency Selection Algorithm

The frequency-selection process refines the minimum-of-maximums algorithm to provide more balanced evaluation when several channels exhibit similar RSSI values.

- Identify all frequencies whose maximum RSSI lies within 2 dB (default) of the best value.
- For each frequency in this group, apply a punishment factor to the opposite-side measurement (controller vs node) proportional to the RSSI difference.
- Recalculate the minimum of maximums on this modified list.
- Select the frequency corresponding to the lowest result.
- If multiple candidates remain equal, select the first in order.

**Note:** If node data is unavailable, the algorithm automatically evaluates the second antenna to determine the best frequency.

### Use Case Example:

If both 2.340 GHz and 2.345 GHz channels show nearly identical RSSI values, the algorithm compares the small differences between the two units' measurements and selects the channel offering the most balanced performance, ensuring symmetrical link quality.

## 10. Extended Scanning Mechanism

In environments with short, sporadic interference bursts (for example Wi-Fi beacons lasting  $\approx 100 \mu\text{s}$  every 150 ms), a standard 4 ms scan may miss activity.

The SkyHopper PRO's Extended Scan Mode increases the sampling window, capturing these transient signals to produce more accurate interference maps.

When Extended Scan Mode is selected in the Web GUI:

- Scan cycle interval: 80 – 100  $\mu\text{s}$
- Sampling time per frequency: 50 – 200 ms (default 150 ms)
- Measured parameters per frequency:
  - RSSI noise floor (per channel)
  - Maximum RSSI (per channel)
  - Average RSSI (for debug)
  - Number of samples collected
  - Reference PER values (RefPER1, RefPER2, RefPER3)
  - Number of samples above RefPER3 threshold

---

These results are used to compute a reference word representing overall channel quality, which the DFS/IAS engine uses to select the optimal frequency.

GUI Control: Under Frequency Configuration → DFS Settings, use the slider to choose Regular Scan or Extended Scan.

Recommendation: Use Extended Scan in ISM bands or congested areas with high Wi-Fi activity.

Caution: Avoid Extended Scan when more than 10 frequencies are configured; the process will require longer scan times and longer disconnection periods during switching.

**Example:**

In a dense industrial zone containing multiple Wi-Fi and Bluetooth sources, Extended Scan Mode allows the SkyHopper PRO to detect micro-bursts of interference and automatically select frequencies that remain clean over time, resulting in a more stable link.



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OFF ☒ ON
Auto Pair Delete Pair

**Dynamic Frequency Selection**

Manual ☒ DFS Manual Frequency

Scan Mode Regular ☒ Extended Not Aged ☒ Age

Range DFS Start  End  Step

Banks DFS Threshold  db

| #  | Primary Frequencies | #  | Secondary Frequencies |
|----|---------------------|----|-----------------------|
| 1  | 2321                | 1  | 2305                  |
| 2  | 2372.5              | 2  | 2350                  |
| 3  | 2696.25             | 3  | 2540                  |
| 4  |                     | 4  | 2420                  |
| 5  |                     | 5  | 2320                  |
| 6  |                     | 6  | 2323.5                |
| 7  |                     | 7  | 2347                  |
| 8  |                     | 8  | 2349.5                |
| 9  |                     | 9  | 2375                  |
| 10 |                     | 10 | 2398.5                |
| 11 |                     | 11 | 2691.25               |

Read Tables From a File
Erase Internal Tables

**Sense and Avoid**

Disable ☒ Enable

Policy

Mild 8 Aggressive

Time Period (sec)

Refresh
Apply

**Figure 14: Frequency Configuration**

| Values Policy scale | percentage of data loss of all sent data (%) |
|---------------------|----------------------------------------------|
| 1                   | 0.5                                          |
| 2                   | 1                                            |
| 3                   | 1.5                                          |
| 4                   | 2                                            |
| 5                   | 2.5                                          |
| 6                   | 3                                            |
| 7                   | 3.5                                          |
| 8                   | 4                                            |
| 9                   | 4.5                                          |
| 10                  | 5                                            |
| 11                  | 7                                            |
| 12                  | 10                                           |
| 13                  | 15                                           |
| 14                  | 20                                           |
| 15                  | 30                                           |

**Table 5 : Policy-to-PER (%) scale for Sense and Avoid**

## 18. ICE Cybersecurity Status

The ICE Cybersecurity Status screen provides an overview of your cybersecurity status. At the top of the page, you'll find an overview of the unit's current security level. A percentage indicates the enabled security features compared to the maximum potential security level achievable with all features enabled.

Additionally, you'll see features that require ICE Professional license. These will be displayed with a small lock icon and a message indicating that a license is required. If you have a license for a feature, the lock icon will not be displayed. Each feature also has an icon indicating whether it is enabled or disabled.

Only when a feature is properly configured and in enable mode, it contributes to the cybersecurity protection score, ensuring the system is secured.



Figure 15: Web GUI "ICE Cybersecurity Status" screen

## 19. Link Monitor

The Link Monitor screen serves as a valuable tool for users to oversee the performance of their RF Data Link. It offers real-time insights into the RSSI and CINR values from both RF channels of the connection.

### **RSSI (Received Signal Strength Indicator):**

RSSI measures the power level received by the radio after it's converted by the antenna. It indicates the relative signal strength in a wireless environment. Higher RSSI values mean stronger signals. Typically, acceptable RSSI values range from 0dBm to -101dBm. A RSSI over -75dBm indicates particularly good link quality. Also, the difference in RSSI values between the two channels should not exceed 5 dB when the units are at least a few meters apart.

### **CINR (Carrier to Interference and Noise Ratio):**

CINR measures signal effectiveness by comparing the carrier signal strength to interference and noise levels. Acceptable values typically range from 8dB to 18dB. Higher CINR values suggest better signal quality.

Users can monitor the Link Monitor graph to ensure that RSSI and CINR values remain within acceptable ranges for optimal performance. The Link Monitor screen displays a graph of the RSSI values over time. SkyHopper PRO utilizes MIMO techniques to enhance RF Data Link performance, resulting in two RSSI and CINR measurements. This enables users to closely monitor the graph and ensure that both RSSI and CINR values stay within acceptable ranges for optimal performance.

| Quality | Color  | RSSI (dBm) | CINR (dB) |
|---------|--------|------------|-----------|
| Poor    | Red    | < -85      | < 7       |
| Fair    | Yellow | -85 to -75 | 7 to 10   |
| Good    | Green  | > -75      | > 10      |

**Table 6 : RSSI color indication for each value range**



Figure 16: Web GUI Link Monitor Page

---

## 20. Throughput

This screen allows users to monitor the data throughput of the SkyHopper PRO system in Mbps. Throughput refers to the amount of data transmitted over the network within a specific time frame and is a crucial measure of the data link's performance. Higher throughput indicates the ability to transfer more data, which is essential for applications requiring real-time data or high-definition video.

Two graphs are displayed on this screen:

- **Total:** Shows the combined throughput of the Controller and Remote units.
- **DL/UL (Download/Upload):** Displays the upload (transmitted Mbps, colored in red) and download (received Mbps, colored in green) throughput for each unit. Note that these values will be opposite at each unit.

Monitoring throughput is important to ensure optimal data link performance and to identify potential issues. Consistently low throughput may indicate interference or other performance-affecting factors. Conversely, if the total graph shows a maximum Mbps value (as configured in the Advanced Configuration), it may indicate that the data link is overloaded, leading to high latency and other performance issues. By regularly monitoring throughput, users can promptly identify and address such issues, ensuring the data link functions as intended.

We recommend setting the throughput to up to 80% of the selected bitrate to maintain optimal performance and avoid potential overloads.



Figure 17:Web GUI Throughput Page

## 21. Scanning Screen

The Scanning Screen is a useful tool for finding the best available frequency for link establishment within the operating range of the SkyHopper. To use this screen, follow these steps:

1. **Start Frequency:** Enter the frequency from which the scanning will begin (in MHz).
2. **End Frequency:** Enter the frequency at which the scanning will end (in MHz).
3. **Frequency Step:** Enter the frequency step (sampling intervals) between each scan within the defined range (in MHz).

**Note:** The "Frequency Step MHz" also defines the estimated sampling time. A higher frequency step or a smaller sampling range will result in a quicker frequency scan.

After setting up the parameters, click the "Scan" button to start the scanning process. The progress bar will show the status of the scan. Once complete, a graph will display the RSSI values (Y-axis) for each frequency over time (X-axis), allowing you to easily identify the best available frequency for link establishment.

**Note:** Applying the search will lead to disconnection of the RF link during spectrum scan, do not perform spectrum scan during operational mode (flying, driving etc...).

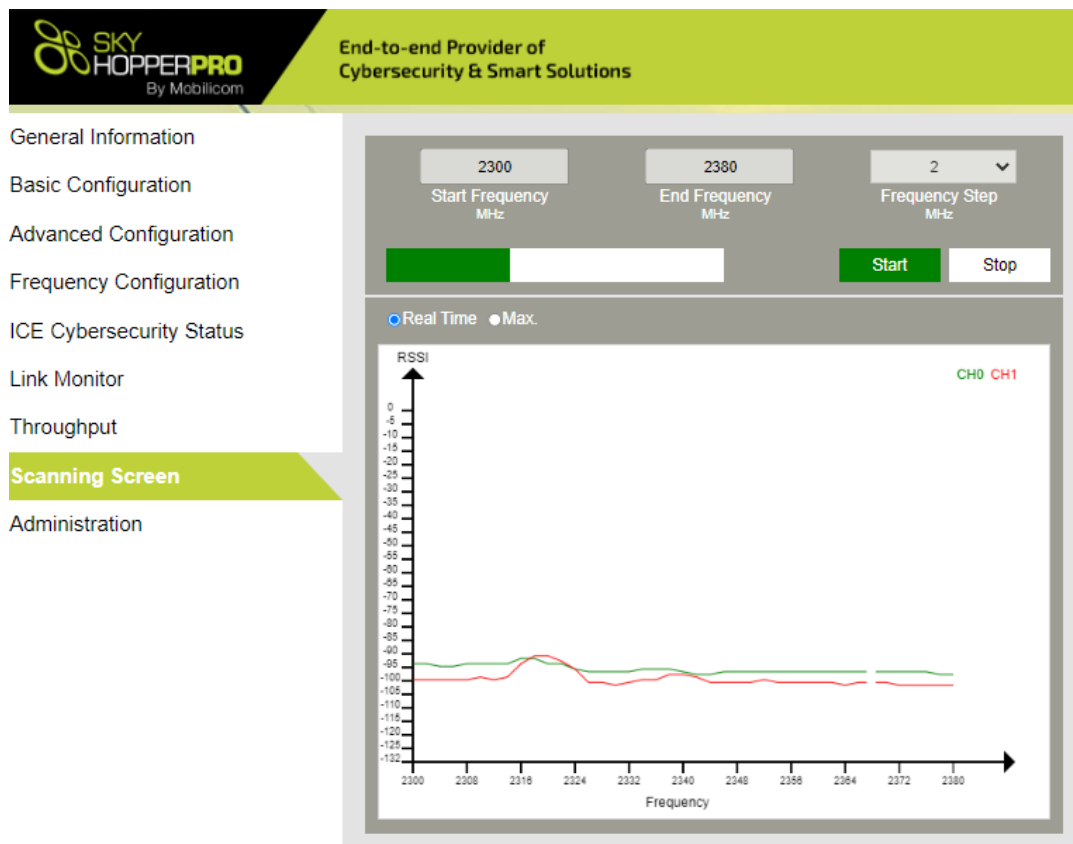


Figure 18: Web GUI Scanning Screen Page

## 22. Administration

The Administration tab provides users with various tools to manage and configure their SkyHopper PRO units.

- **Reboot the Unit:** This option allows users to reboot the SkyHopper PRO unit. It can be useful when troubleshooting or after performing firmware upgrades or configuration changes.
- **Restoring the Unit to Factory Default:** This option allows users to reset the SkyHopper PRO unit to its original factory settings. All custom configurations and settings will be lost, so it is recommended to use this option only when necessary.
- **Uploading Configuration Files:** Users can upload configuration files to the SkyHopper PRO unit. This is useful when deploying multiple units with the same configuration settings, as it can save time and effort.
- **Uploading licensing:** Users can upload licensing files to the SkyHopper PRO unit. This is necessary to enable specific features that require a license.
- **Uploading policies (radio):** Users can upload policies file to the SkyHopper PRO unit. This is necessary to enable specific radio policies that require.
- **Swapping from Bank 1 to Bank 0:** SkyHopper PRO units have two firmware banks, Bank 0 and Bank 1. This option allows users to switch between the two banks. Both PTP and Mesh software versions are supported simultaneously (only from version 1.43.4 and above for PTP and from 3.18.4 and above for Mesh). This feature allows you to save two independent firmware versions for redundancy or to switch between PTP and Mesh configurations.
- **Upgrading the Unit:** SkyHopper PRO units have two firmware banks, Bank 0 and Bank 1. This option allows users to switch between the two banks. Both PTP and Mesh software versions are supported simultaneously (only from version 1.43.4 and above for PTP and from 3.18.4 and above for Mesh).

All software files released by the Mobilicom Sky Hopper development team follow a specific naming convention. The naming format is structured as follows:

*MC-SW-[Topology]-[HW Revision][Type]-[Version Number]-[Client String]\**

**\*Important Note:** The Client String is optional.

### Naming Convention Breakdown

- MC: Mobilicom vendor prefix
- SW: Software category
- Topology: Applicable topology (PTP or MTM)
- HW Revision: Digital card hardware version (D, E, or X for "don't care")
- Type: Software type, specified as:
  - B: Boot update file only
  - P: Package file, including Boot update, Software update, and Configuration file (if it does not contain a client string, it will restore factory default)
  - N: Boot and Software update
  - U: Update only

- F: Update and Configuration (if it does not contain a client string, it will restore factory default)
  - C: Configuration file (if it does not contain a client string, it will restore factory default)
- Version Number: Version release number
  - Versions starting with 1.x.x (e.g., 1.43.6) are always PTP
  - Versions starting with 3.x.x (e.g., 3.18.5) are always MTM
- Client String: (Optional) Used for customized configurations or full versions with customized configurations

#### **Naming Convention Example**

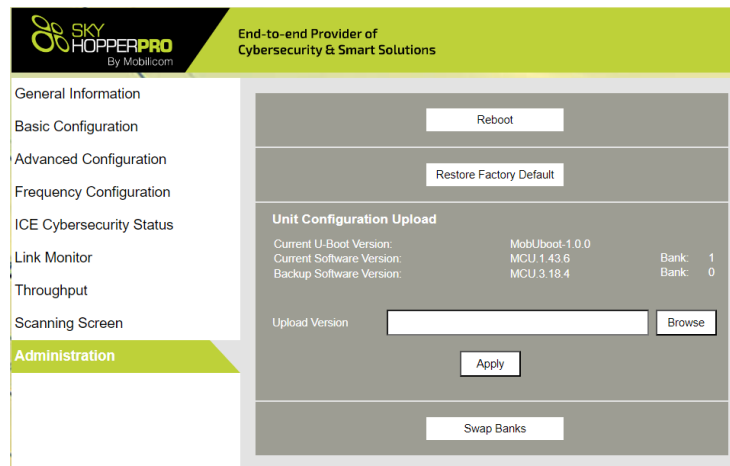
- MC-SW-PTP-DP-1.41.19:
  - MC: Mobilicom
  - SW: Software
  - PTP: Point-to-Point topology
  - D: Hardware revision D
  - P: Package file, including Boot update, Software update, and Configuration file (restores factory default if no client string is included)
  - 1.41.19: Version number
- MC-SW-MTM-EXN-3.18.5:
  - MC: Mobilicom
  - SW: Software
  - MTM: Multi-Topology Mesh
  - E: Hardware revision E
  - X: Any hardware revision
  - N: Boot and Software update
  - 3.18.5: Version number

\*Note: This naming convention applies only to the image package file names.

Before upgrading your SkyHopper, please consult the relevant release notes to verify the necessary steps, version type, and version level required.

#### **Note:**

- When upgrading firmware, ensure a stable power source. The process may take several minutes, so avoid power cycling the units during this procedure.
- Upgrade the firmware for each unit individually, remote first, while no video stream active in the link during the upgrade.



**Figure 19: Web GUI Administration Page**

## 23. Connecting the Antennas

Connecting the antennas to the SkyHopper PRO is a crucial step in setting up the device. Here are some additional details to keep in mind:

1. **Antenna Placement:** Antennas should be placed as high and clear as possible to minimize obstructions and maximize signal strength.
2. **Antenna Selection:** The choice of antenna will depend on the specific application and requirements. Different types of antennas, such as directional or omnidirectional, may be more suitable for certain applications.
3. **Cable Selection:** The cables used to connect the antennas should be chosen carefully, considering factors such as attenuation and flexibility. Lower attenuation cables will provide better signal strength.
4. **Connector Type:** The cables used to connect the antennas should have an MMCX Male connector on the SkyHopper PRO end and a connector that matches the antenna's connector on the other end.
5. **Antenna Orientation:** The orientation of the antenna can affect the signal strength and quality. It is important to refer to the manufacturer's specifications for the correct orientation and positioning of the antenna.
6. **Antenna impedance:** The antenna impedance should be 50 ohms. This impedance is standard for most RF applications and ensures optimal performance and signal integrity.

By following these guidelines, you can ensure a reliable and efficient connection between the SkyHopper PROV and the antennas.

---

## 24. Connecting Power

**CAUTION:** It is important to note that powering up a SkyHopper PRO without connecting the antennas can cause irreparable damage to the unit.

To connect power to the SkyHopper PRO, follow these steps:

1. **Identify the Power Cables:** Identify the power cables that came with the unit. These should be included in the package or purchased separately.
2. **Connect the Power Cable:** Connect one end of the power cable to the power input on the SkyHopper PRO.
3. **Connect to Power Source:** Connect the other end of the power cable to a power source, such as a battery or power supply (7-24V). Ensure to use the appropriate voltage and current specifications.
4. **Power On:** Once the power is connected, the unit should turn on automatically. If it does not turn on, check the power source and cable connections.

---

## 25. API

Mobilicom's API is based on its proprietary MCIP Protocol. The MCIP protocol is a textual communication protocol that is implemented in Mobilicom's MCU and SkyHopper units for configuration and monitoring of the units. The protocol provides remote control capability to applications such as Mobilicom's WebGUI and customer's Command and Control Systems.

The protocol is built from an MCIP message header that can be concatenated with more content.

The MCIP protocol is ASCII text-based (utilizing easy debug sessions). The protocol supports the transport of several data types.

For more details, please contact Mobilicom support.

## Appendix A - Mechanical Drawings

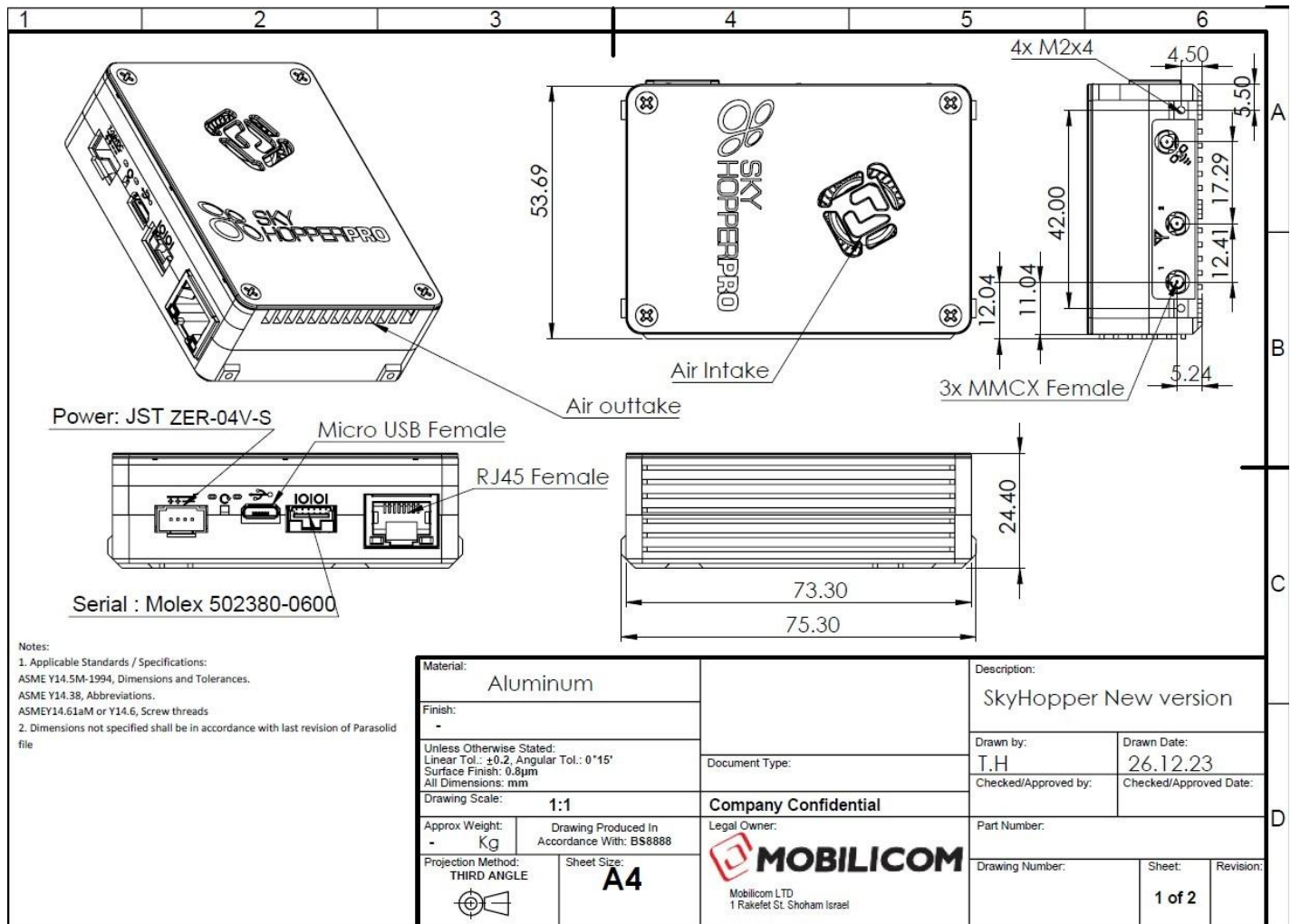
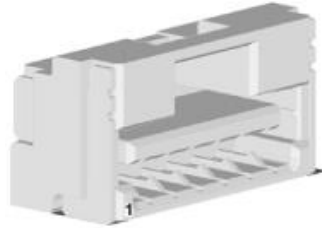


Figure 20: Sky Hopper PRO Mechanical Drawing

## Appendix B - Connectors' Pinout

RS-232/1PPS



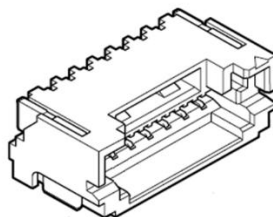
**Figure 21 RS232/1PPS connector (IOIOI) (SkyHopper PRO side)**

| Housing P/N: 502380-0600<br>Pins P/N: 502381-0000 | Name      | Description<br>(DB9 cable for RS232) |
|---------------------------------------------------|-----------|--------------------------------------|
| 1                                                 | N/A       |                                      |
| 2                                                 | RS232 out | Pin 2                                |
| 3                                                 | N/A       |                                      |
| 4                                                 | RS232 in  | Pin 3                                |
| 5                                                 | GND       |                                      |
| 6                                                 | 1PPS In   |                                      |

**Table 7: Connectors' Pinout Serial**

Power Connector

Note: The voltage range for the Sky Hopper PRO is 7.4-24V DC.



**Figure 22: Power connector (SkyHopper PRO side)**

| Housing P/N: ZER-04V-S, Pins P/N: SZE-002T-P0.3 | Description |
|-------------------------------------------------|-------------|
| 1                                               | GND (-VDC)  |
| 2                                               | GND (-VDC)  |
| 3                                               | Vin+ (+VDC) |
| 4                                               | Vin+ (+VDC) |

**Table 8: Connector Pinout Power**

## Appendix C – USB Eth connection

The Micro USB port in SkyHopper PRO can be used for ETH connectivity, allowing ETH over a standard Micro USB to USB cable.

For windows-based systems a Driver must be installed to allow ETH over USB connection and configuration.

Please contact Mobilicom for the driver files and configuration instructions.



**Figure 23: USB Ethernet connection**

## Appendix D – Troubleshooting

| Symptom                                       | Possible Cause                                                                                                                                                                                                                                                                                                     | Action Needed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LED is off                                    | Not connected to power source or battery is empty                                                                                                                                                                                                                                                                  | <p>Verify power cable properly connected to the Sky Hopper PRO power connector.</p> <p>Verify that the battery or power supply is connected to the Sky Hopper PRO power cable and to a power outlet.</p> <p>Replace the battery or check the outlet.</p> <p>Replace the power cable.</p>                                                                                                                                                                                                                                                                                            |
| LED light is always on                        | PTP – Unit is in boot mode                                                                                                                                                                                                                                                                                         | <p>Check the power source.</p> <p>Reboot the SkyHopper PRO.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Cannot connect to SkyHopper PRO using Web GUI | <p>The PC is not configured with a correct static IP.</p> <p>Unit is not powered up.</p> <p>Ethernet cable is not connected, or data cable is damaged.</p> <p>Firewall is blocking traffic.</p> <p>ETH port set to “auto negotiation off” with 1GE.</p> <p>Trying to connect to remote unit when link is down.</p> | <p>Verify your PC is set with a static IP in the subnet of 192.168.131.xxx</p> <p>Verify that you use https:// before the unit IP.</p> <p>Check that the ETH port is set to “auto negotiation on.”</p> <p>Verify the unit is powered up.</p> <p>Verify that the PC ETH port status is not unplugged or disabled.</p> <p>Disable the PC firewall.</p> <p>Verify ETH cable is connected to the PCs ETH port and to the Sky Hopper PROV MAIN connector.</p> <p>If needed, replace the cables.</p> <p>Ping the unit.</p> <p>Try to connect to the local unit and verify link is up.</p> |

|                                             |                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                  |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No link between SkyHopper PRO units         | <p>Configuration is not the same on all the SkyHopper PRO units.</p> <p>PTP – both units set as Remote or both as Controller.</p> <p>The units are not within reception range</p> | <p>Connect to each unit and verify they have the same configuration.</p> <p>Verify one unit is set as a Remote and the other as a Controller.</p> <p>Verify that TX power is set to the maximum.</p> <p>Verify the antennas fit the Frequency used (check the antenna spec).</p> |
| Data with high delay and disconnecting data | <p>Trying to send a higher amount of data than the allocated bandwidth sent to the unit</p>                                                                                       | <p>Verify that the data bandwidth sent to the unit from the connected device(s) does not exceed the allocated bandwidth (in its configuration).</p> <p>Decrease the data received by the data source or change the allocated bandwidth via the Web GUI.</p>                      |

**Table 9: Troubleshooting**

For technical support for Sky Hopper PRO, please contact:

[support@mobilicom.com](mailto:support@mobilicom.com)